

Shivangi Kumar

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EDUCATION

Columbia University

New York, NY

Master of Science in Computer Science (Specialization in AI/ML), GPA: 4.1/4.0

Dec 2026

NLP, Generative AI, Deep Learning for Computer Vision, Large-Scale Distributed Systems, Advanced Big Data & AI

Indian Institute of Technology Guwahati

Guwahati, India

Bachelor of Science in Computer Science, GPA: 8.06/10.0

May 2022

Algorithms, Data Structures, Machine Learning, Deep Learning, Operating Systems, Computer Networks, DBMS

PROFESSIONAL EXPERIENCE

Amazon

Seattle, WA

Software Development Engineer Intern - Subscribe & Save Customer Experience

Jun 2026 – Aug 2026

- Owned end-to-end delivery of **Subscribe & Save Discover Page uplifts**, improving product discovery, recommendations, and checkout experience **across 7 marketplaces**
- Increased product visibility by 40%** per mobile screen through a compact carousel redesign, validated via an A/B experiment rolled out to 50% of global Subscribe & Save traffic
- Diversified user recommendations by 3x**, adding anti-clustering constraint to the widget ranker, preventing similar widgets from clustering
- Extended the Discover Page to support the one-now-more-later order type, **cutting time to first delivery by up to 85%**
- Leveraged GenAI agents (Claude Code, Kiro) across design, implementation, testing, and ops to complete three assigned milestones ahead of schedule and deliver more beyond scope

Goldman Sachs

Bengaluru, India

Software Engineer

Jul 2022 – Aug 2025

- Led the redesign of client onboarding platform for the Futures & Options trading business to remove a scaling bottleneck; the new architecture supports **10x client capacity** and **cut onboarding time by 30%**
- Upgraded the order execution engine to **support 2x instrument types** on a latency-sensitive path, **increasing throughput by 65%**
- Built a platform automating access management across **20K+ servers** and real-time monitoring for **25K+ servers** backing critical financial infrastructure, **eliminating 90% of manual approval** and cutting incident response times by **25%**

Summer Intern

May 2021 – Jul 2021

- Built a distributed monitoring system covering 20K+ servers globally, **boosting ops efficiency by 40%**; received a pre-placement offer for a full-time role

PROJECTS

Self-Learning Speech-to-Text System with Autonomous Error Correction

Dec 2025

- Developed a **self-adapting speech-to-text pipeline** that detects its own transcription errors (wav2vec2), corrects them with an LLM (Llama 3.2 3B via Ollama), and fine-tunes (LoRA) on the corrections, **improving word error rate by 15.6%**
- Designed an **adaptive scheduler** that triggers fine-tuning only when accuracy gains justify the compute cost, **cutting unnecessary retraining runs by 40-60%**

Rx-AI – Multimodal Clinical Intake System

2026

- Built a **multimodal patient check-in agentic pipeline** (FastAPI + React) that generates personalized questionnaires via a CrewAI three-agent pipeline over Vertex AI Gemini, with **integrated speech-to-text, text-to-speech, and image analysis**
- Benchmarked the multi-agent pipeline against a single-pass baseline with **JSONL eval logging**, surfacing a **quality-vs-latency tradeoff** that guided the production path selection

AI Wardrobe Planner – Vision-Based Outfit Recommender

2026

- Built a **computer-vision pipeline** (YOLOv8 + EfficientSAM + CLIP) that auto-extracts a **structured digital wardrobe** from user photos/videos, served by a **FastAPI + PostgreSQL** backend deployed via Docker
- Integrated **calendar, location, and weather signals** into a **context-aware recommender** for event and weather appropriate outfits

TECHNICAL SKILLS

- Languages:** Java, Python, C++, TypeScript/JavaScript
- ML / AI:** PyTorch, TensorFlow, Keras, NLP, LLM fine-tuning (LoRA), wav2vec2, Graph Neural Networks (RGCN), GradCAM, Ollama
- Backend & Systems:** Distributed Systems, Microservices, REST APIs, Kafka, Prometheus, Linux
- Cloud & Tools:** AWS (Lambda, SQS, DynamoDB, CloudWatch), Docker, Kubernetes, Terraform, Git, CI/CD, Claude Code
- Databases:** SQL, DynamoDB, Firebase